b) $\text{Var}(X) = E[X^2] - (E[X])^2$

$= E[X^2] - q$

$= 12 - 9 = 3$

$E[X^2] = \sum_{k=0}^{\infty} k^2 \cdot \left(e^{-3} \cdot \frac{3^k}{k!}\right)$

$= e^{-3} \sum_{k=0}^{\infty} k^2 \left(\frac{3^k}{k(k-1)!}\right)$

$= e^{-3} \sum_{k=0}^{\infty} (k(k-1)+k)\left(\frac{3^k}{k!}\right)$

$= e^{-3}\left(\sum_{k=2}^{\infty} \frac{k(k-1)3^k}{k(k-1)(k-2)!} + \sum_{k=1}^{\infty} \frac{k \cdot 3^k}{k!}\right)$ $\quad k(k-1)(k-2)!$

$= e^{-3}\left(9\sum_{k=0}^{\infty} \frac{3^{k-2}}{(k-2)!} + 3\sum_{k=0}^{\infty} \frac{3^{k-1}}{(k-1)!}\right)$

$= 3e^{-3}\left(3\sum_{k=2}^{\infty} \frac{3^{k-2}}{(k-2)!} + \sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!}\right)$

$= 3e^{-3}\left(3e^3 + e^3\right)$

$= 12$

Q2: $P(A)=0.5$ $P(B)=0.3$ $P(A|B)=0.6$

Bayes: $P(B|A) = \frac{P(B) \cdot P(A|B)}{P(A)}$

$= \frac{0.3 * 0.6}{0.5}$

$= 1.8/5 = 0.36$

a) $P(A \cap B) = P(A|B) \cdot P(B)$

$= 0.6 * 0.3 = 0.18$

b) $P(B|A) = 0.36$

c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18$

$= 0.62$

Q3: a) $P(H,H,T,H,T|\theta)$ MLE

$L(\theta) = \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta)$

$= \theta^3 (1-\theta)^2$

$\ln L(\theta) = \ln(\theta^3 (1-\theta)^2)$

$= \ln \theta^3 + \ln(1-\theta)^2$

$= 3 \ln \theta + 2 \ln(1-\theta)$

$\frac{d(\ln L(\theta))}{d\theta} = \frac{3}{\theta} + \frac{-2}{1-\theta} = 0$

$= 3(1-\theta) - 2\theta = 0$

$= 3 - 3\theta - 2\theta$

$= 3 - 5\theta \rightarrow \theta = \frac{3}{5}$

b) $\theta^3 (1-\theta)^2 \cdot 2\theta$

$= 2\theta^4 (1-\theta)^2$

$\ln\left(2\theta^4 (1-\theta)^2\right)$

$= \ln 2 + 4 \ln(\theta) + 2 \ln(1-\theta)$

$\frac{d(\ldots)}{d\theta} = \frac{4}{\theta} + \frac{-2}{1-\theta} = 0$

$= 4(1-\theta) - 2\theta = 0$

$= 4 - 4\theta - 2\theta \rightarrow \theta = \frac{2}{3}$

Q4: a) $f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\sigma^2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$

$L(\mu, \sigma^2) = \prod_{i=1}^{n} f(x_i \mid \mu, \sigma^2)$

$= \left(\frac{1}{\sqrt{2\sigma^2\pi}}\right)^n e^{-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}}$

$\ln L = -\frac{n}{2} \ln(2\sigma^2\pi) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$